

Vutshila Mashimbyi

BSc Computer Science

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About Me

Studied computer science at the University of the Witwatersrand, with certification in Full Stack Development from the FNB App Academy. Passionate about full-stack development, data science, AI/ML, and cybersecurity. Combining strong theoretical foundations with practical development skills to build innovative digital solutions. Equipped with comprehensive knowledge in algorithms, data structures, software design, and modern development practices through rigorous academic training.

Education

BSc Computer Science

University of the Witwatersrand

- Completed degree with comprehensive education in algorithms, data structures, software engineering, and database management
- Recognized as top achiever in first year residence academic rankings
- Participant in Wits Maths Competition

Certificate in Full Stack Development

FNB App Academy (2025)

- Earned 32 credits in intensive full-stack development program
- Key coursework: App Strategies, GitHub Collaboration, SDLC with HTML
- UX Design Principles, API Development, Data Management
- Business Development, Backend Development (API Building)
- AI in Development, App Marketing Strategies
- Certificate Number: A0TYA07Z025 (Attached)

Academic Foundation

- **Basic Computer Organisation** : Computer architecture, binary systems, data representation, logic gates, low-level programming (Pep-8), operating systems principles
- **Discrete Computational Structures** : Logic, proofs, algorithm complexity, number theory, cryptography, induction, recursion, combinatorics, finite-state machines, Turing machines
- **Data Structures & Algorithms** : Arrays, linked lists, stacks, queues, trees (binary search trees, AVL trees), heaps, sorting algorithms, graph traversals (BFS/DFS), time/space complexity analysis in C++
- **Computer Networks** : Layered network architecture (Application, Transport, Network, Link layers), TCP/UDP protocols, IP addressing, routing algorithms, DNS, HTTP, network traffic analysis
- **Database Fundamentals** : Relational database design, Entity-Relationship modelling, SQL programming, normalization, relational algebra/calculus, transaction management
- **Mobile Computing** : Android application development, Java programming, object-oriented

design, Android architecture (Activities, Intents, Views), UI design, sensor integration, API consumption

- **Advanced Analysis of Algorithms** : Greedy algorithms, dynamic programming, complexity classes (P, NP, NP-Completeness), AI algorithms, computational geometry, advanced data structures
- **Parallel Computing** : Parallel computer architectures, performance modeling, cooperative problem-solving, multi-processing techniques, parallel algorithm design
- **Formal Languages & Automata** : Turing machines, decidability, computational complexity, polynomial-time reductions, Gödel's incompleteness theorems
- **Software Design & Engineering** : Software development methodologies (Agile/Scrum, Waterfall), requirements gathering, object-oriented design, software architecture patterns, design patterns, SOLID principles, software testing
- **Machine Learning** : Linear regression, logistic regression, Naïve Bayes, decision trees, clustering, neural networks, reinforcement learning, practical ML application
- **Operating Systems** : Processes and inter-process communication, multi-threaded programming, memory allocation and management, resource allocation and scheduling, file systems and persistent storage, protection and security mechanisms
- **Computer Graphics and Visualization** : Computer graphics fundamentals, 3D modelling techniques, rendering processes and algorithms, animation principles, data visualization methods, computational geometry algorithms, image synthesis and manipulation

Professional Experience

Computer Science Tutor

University of the Witwatersrand

- Tutored first and second year students in programming fundamentals, data structures, and algorithms
- Conducted tutorial sessions and assisted with assignments
- Helped students develop computational problem-solving skills

Scrum Master & Team Lead (Skills)

Constitutional Archive Project

- Led development team in creating full-stack constitutional archive management system
- Implemented Agile methodologies and managed sprints
- Coordinated between stakeholders and developers

Maths & Science Ambassador

Winfred Nomzamo Mandela High School

- Facilitated weekly quizzes to promote peer learning
- Recognized and mentored top academic performers

Technical Skills

- **Programming Languages** : Java, Python, JavaScript, C++, SQL, HTML/CSS
- **Web Development** : React/Next.js, Node.js, Three.js, Firebase, REST APIs
- **Development Tools** : Git/GitHub, Docker, Linux, Azure, VS Code
- **Data Science & AI/ML** : Pandas, NumPy, Scikit-learn, TensorFlow, Data Visualization
- **Database Systems** : MySQL, PostgreSQL, Database Design, Query Optimization

- **Mobile Development** : Android SDK, Java, Mobile UI/UX Design
- **Software Engineering** : Agile/Scrum, Software Architecture, Design Patterns, Testing

Professional Skills

- Continuous learning and skill development (demonstrated by FNB certification)
- Critical thinking and complex problem solving
- Project management and team leadership
- Effective collaboration and communication (GitHub Collaboration certified)
- Analytical approach to technical challenges
- Algorithm design and optimization
- System architecture and design patterns

Key Projects

Campus Dining Services App (React, Node.js, Firebase)

Full-featured application for campus dining with real-time menus and reservations

[Live Demo](#) — [GitHub](#)

Nitro Assault Racing Game (Three.js, JavaScript)

Time-trial racing game with physics and multiple camera perspectives

[Play Demo](#) — [GitHub](#)

Constitutional Archive Management System (Vanilla JS, Node.js, Azure)

Full-stack web application for managing historical constitutional data with NLP search

[Live Demo](#) — [GitHub](#)